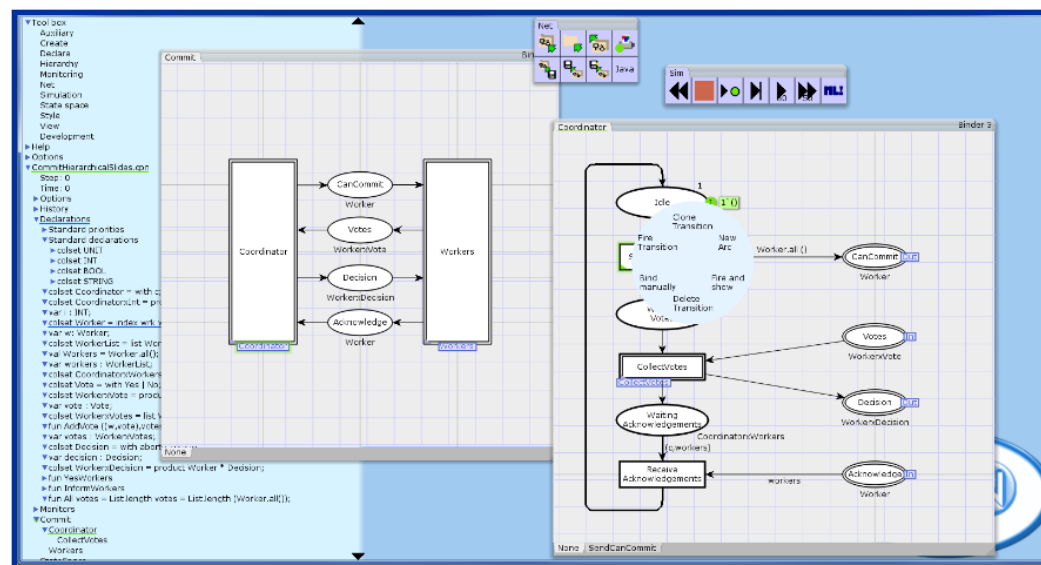




Western Norway
University of
Applied Sciences

My background

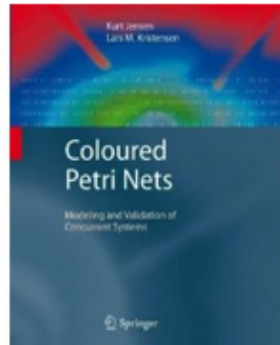


- **2000:** PhD from the **CPN research centre** at Aarhus University (DK) on Coloured Petri Nets and software verification
- **2000-2002:** Post-doctoral researcher at University of South Australia | Australian Defence and Technology Organisation
 - Software tool support for military command and control
 - Modelling and implementation of real-time avionics mission systems
- **2002-2009:** Associate professor at Aarhus University
 - Capacity planning for web servers with Hewlett-Packard
 - Development of mobile ad-hoc network protocols and IPv6 with Ericsson Telebit
- **Since 2009:** Professor of computer science and software engineering at Western Norway University of Applied Sciences
 - Establishment of a PhD programme in Computer Science and a master programme in Applied Computer Science and Engineering
 - Teaching and research: network technology and distributed systems middleware, IoT and cloud software systems, model-driven software engineering and verification

CPN textbook

Coloured Petri Nets: Modelling and Validation of Concurrent Systems

Welcome to the homepage of the CPN Book



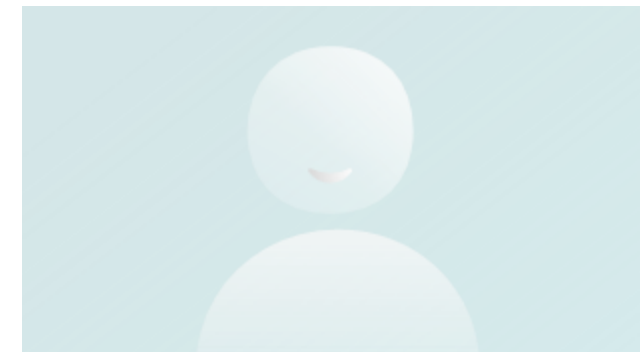
Springer, July 2009 - available via: [Springer](#) [amazon.co.uk](#) [amazon.com](#)

Authors

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Aarhus University, Denmark	Western Norway University of Applied Sciences

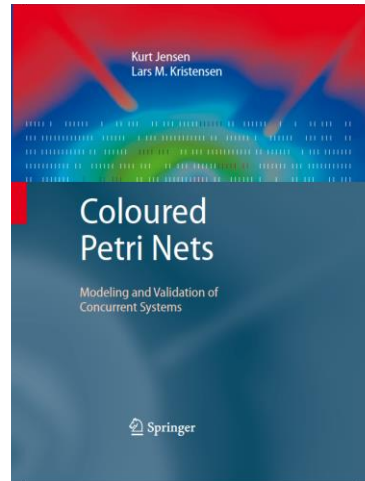
Introduction

Coloured Petri Nets (CP-nets or CPNs) is a language for modelling and validation of concurrent and distributed systems and other systems in which concurrency, synchronisation, and communication plays a major role. The CPN textbook



- **K. Jensen and L.M. Kristensen. Coloured Petri Nets: Modelling and Validation of Concurrent Systems, Springer, 2009.**
- **Book website: www.cpnbook.org**

Resources



K. Jensen and L.M. Kristensen.
Coloured Petri Nets: Modelling and Validation of Concurrent Systems,
Springer, 2009.

www.cpnbook.org

Practical use of CPN Tools is extensively documented at
www.cpntools.org

Coloured Petri Nets: A Graphical Language for Formal Modelling and Validation of Concurrent Systems

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ABSTRACT

Coloured Petri Nets (CPNs) combine Petri nets with a powerful programming language to obtain a suitable modelling language for concurrent systems. This paper provides the formal basis for the modelling, simulation, and validation of CPNs. A programming language provides the primitives for modelling the behaviour and control of concurrent systems. The CPN modelling language, and finally, a set of rules for the simulation and validation of CPNs.

Coloured Petri Nets and CPN Tools for modelling and validation of concurrent systems

Coloured Petri Nets (CPNs) are a graphical modelling language for concurrent systems based on Petri Nets and the functional programming language Standard ML. CPN Tools provide the primitives for modelling, simulation, and validation of CPNs. The CPN modelling language, and finally, a set of rules for the simulation and validation of CPNs.

Implementing Coloured Petri Nets Using a Functional Programming Language

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Abstract. Coloured Petri Nets (CPNs) are a graphical modelling language for concurrent systems based on Petri Nets and the functional programming language Standard ML. CPN Tools provide the primitives for modelling, simulation, and validation of CPNs. The CPN modelling language, and finally, a set of rules for the simulation and validation of CPNs.

Keywords: Coloured Petri Nets, Functional Programming Language, Simulation, Validation, Performance Analysis, Concurrent Systems, Formal Verification. In the modelling language, Standard ML, we model the state of the system and the CPN. The CPN modelling language, and finally, a set of rules for the simulation and validation of CPNs.

1. INTRODUCTION

The paper presents an overview of how functional programming and Standard ML are applied in the CPN modelling language. The CPN modelling language is a graphical modelling language for concurrent systems based on Petri Nets and the functional programming language Standard ML. CPN Tools provide the primitives for modelling, simulation, and validation of CPNs. The CPN modelling language, and finally, a set of rules for the simulation and validation of CPNs.

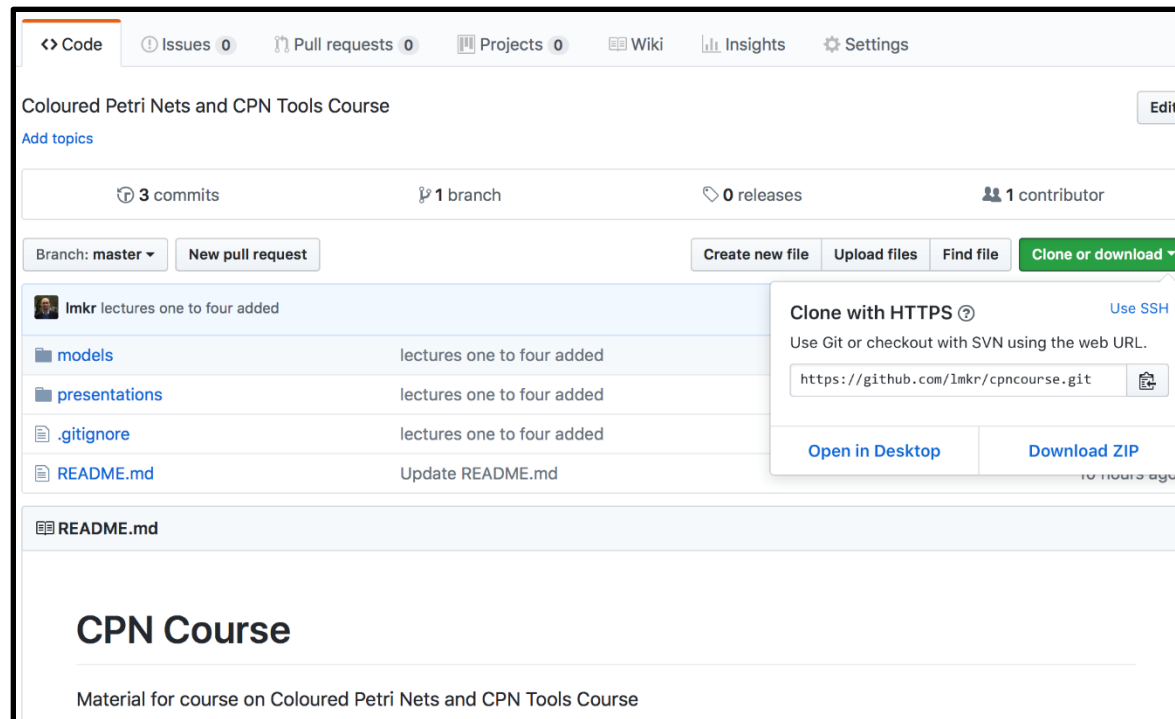
■ Research papers on Coloured Petri Nets (CPNs)

- K. Jensen and L.M. Kristensen. Coloured Petri Nets: A Graphical Language for Modelling and Validation of Concurrent Systems. Communications of the ACM, Vol. 58, No. 6, pp. 61-70, 2015.
- K. Jensen, L.M. Kristensen, L. Wells. Coloured Petri Nets and CPN Tools for Modelling and Validation of Concurrent Systems. Intl. Journal on Software Tools for Technology Transfer, Vol. 9, pp. 213-254, Springer, 2007.
- L.M. Kristensen and S. Christensen. Implementing Coloured Petri Nets using a Functional Programming Language. In Higher-order and Symbolic Computation, Vol. 17, pp. 207-243, 2004.

Course material

- Slides, models, and papers are available via the github repository at

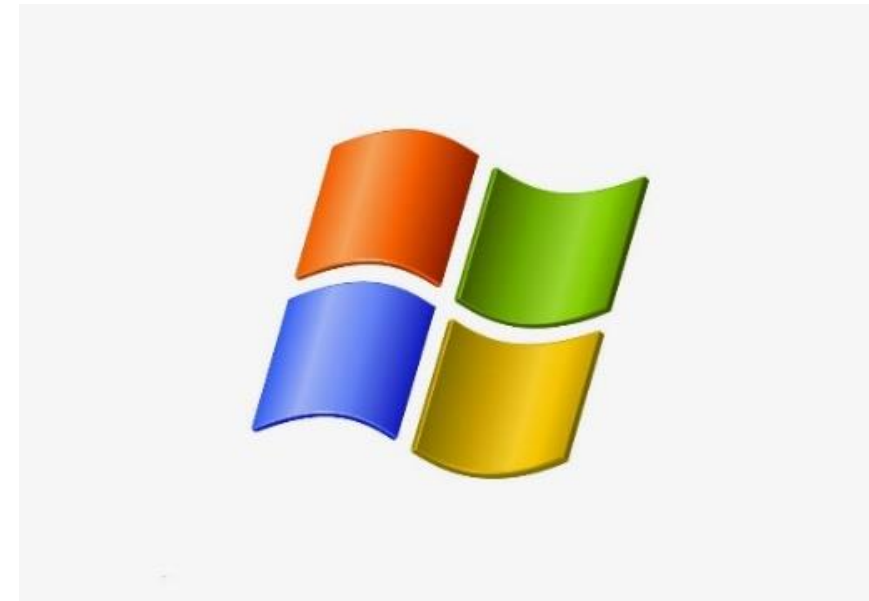
<https://github.com/Imkr/cpncourse>



Clone the git-repository or download as a zip-file

CPN Tools installation

- CPN Tools can be downloaded and installed via www.cpntools.org



- Some installations of Windows requires the application to be run as administrator

Outline



- **Motivation and overview of Coloured Petri Nets (CPNs)**
- **Background** on Coloured Petri Nets:
From Place/Transition Nets to Coloured Petri Nets?
- The **syntax** and **semantics** of the basic constructs of the Coloured Petri Nets (CPNs) modelling language
- Modules for **hierarchical structuring** of CPN models of large systems

